

Children's evaluation of a virtual reality boating safety education resource

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ABSTRACT

Research on designing and using immersive virtual reality with children for educative purposes is an emerging field. It is important to deploy participatory design techniques to highlight the perspectives of children so that the design of content, characters, interactive elements and narrative resonate with young learners. This paper reports on an evaluation of a proof-of-concept 360-degree learning resource designed to educate children about boating safety. The resource was developed as part of a government service safety initiative to be integrated into a short in-school boating safety session facilitated by maritime safety officers. A mixed methodology was used during a focus group with primary (elementary) school aged children. Findings were that children wanted to experience the learning resource via a virtual reality headset with their Systems Usability Scale results being positive. However, most had difficulty adjusting the headset for visual clarity and a minority managed to experience all 360-degree scenes and their embedded content. Children enjoyed the interactive nature of the resource, felt transported to the boating location and liked human and animated characters in the story. The evaluation was useful in clarifying aspects of the learning resource that were valued by children and it raised questions about whether the resource might best be used during the short in-school boating safety sessions in computer or mobile device screen mode rather than via a virtual reality headset.

Keywords: Human computer interaction (HCI), HCI design and evaluation methods, Field studies, Virtual reality, Children, Education.

Index Terms: Human-centered computing — Human computer interaction (HCI) — Interaction paradigms — Virtual reality; Social and professional topics — User characteristics — Age — Children; Human centered computing — Human computer interaction (HCI) — HCI design and evaluation methods — Field studies.

1 INTRODUCTION

The theoretical basis for the learning affordances of virtual reality (VR) have been identified for several decades [1, 2, 3]. Some of the affordances or properties of VR that might enhance learning include: the ability of the user to view and interact with objects, beings and environments from a first-person perspective; the option of practicing real life procedures in safe or low stakes virtual places; the manipulation of size and scale to enhance understanding; the power to transport users to places and times that are impossible or difficult to experience in real life; and the transformation of fairly abstract ideas into observable representations that can be manipulated to increase understanding. While the theoretical basis of VR for education is reasonably

defined, far less is known about designing effective VR learning experiences for children and pedagogically deploying the technology in natural educational settings. This is especially pertinent for content that can be experienced on screen-based computing device such as a tablets or smart phones and via a head mounted display (popularly known as a VR headset) [4, 5]. Research on immersive VR for education (that is VR mediated by a headset) is a young field as commercial availability of relatively affordable VR headsets with appropriate learning content only dates from 2014 - 2016 (depending on hardware and software developments to enable three or six degree-of-freedom functionality).

Understanding children's (end-user) interactions with and perspectives on VR educational experiences is important if the technology is to be appropriately designed and effectively implemented for educational purposes [6]. This paper reports on an evaluation of a pilot 360-degree VR application designed to teach children about boating safety. The interactive application, which could be experienced on a desktop, tablet, or mobile computer device such as a smart phone which could also be inserted into a VR headset. It was developed and piloted as part of an education program conducted in schools by maritime safety officers employed by state agency Transport for New South Wales (NSW). Situated within the intersection of educational research and participatory design for human-computer interaction, this paper reports on a mixed method study that highlighted children's perspectives on the boating safety resource to better understand how such VR applications might be designed and integrated into community education programs.

2 PRIOR RESEARCH

A recent systematic review of research on the topic VR in education [5] identified 46 journal articles published between 2009 to mid-2020. Twenty-one studies were K-12 with some suggesting that the technology could be used with children and young people to develop an understanding of complex material, metacognition, creativity and problem-solving. Only 2 of the 21 studies in the review indicated a collaborative or participatory approach with teachers or students.

Some research has investigated the effects of learning the same material with VR versus other media, and this shows equivocal results [7, 8]. One particularly relevant study [9] that compared a 360-degree water (beach) safety resource with the same material delivered via video or poster found exposure to all media increased children's knowledge and retention of knowledge about beach safety. The video or poster exposure had slightly higher knowledge acquisition and retention which the author postulate was due to children interacting with a water safety expert while learning with this media. They conclude that the independent learning that the 360-degree video group of children undertook demonstrated the media could be effective for beach safety education.

Participatory design is an approach where the designers of computing artefacts invite future users to participate in some or all phases of the design process [10]. There is a substantive body of research on participatory design in the human-computer

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interaction including the field of child-computer interaction [11]. Cumbo, Erikson and Iverson state that a participatory approach with children generates more pertinent technological outcomes and ethically empowers children by giving them a voice in decisions so that they can develop competencies in design [12]. While most participatory design is adult-led [12], some researchers suggest that children can play some or all of these roles of user, tester, informant and partner in the design process [13]. For the research reported in this paper, children undertook the roles of user and tester (trying out the 360-degree boating safety resource to offer feedback on its aesthetics, useability, modes of interactivity and game mechanics, and educational foci).

3 THE VR BOATING SAFETY RESOURCE

Understanding the context for the development of the 360-degree VR boating resource for children is important. Australia is an island continent with the bulk of its population living on the coast. It also has an abundance of interior waterways (rivers, lakes, billabongs, and dams). In the Australian state of NSW, which was the setting for the school-based maritime education program that the VR resource is being designed for, there are approximately 550,000 people who hold a licence to drive a powered vessel and nearly 250,000 registered boats in the state, with 1 in 5 households owning a boat or watercraft [14]. The proximity and popularity of waterways for leisure has led to significant government and community investment in water safety including boating safety programs. While many children enjoy boating as part of leisure, children from the age of 12 can undertake boat license requirements in NSW.

Maritime NSW safety officers were interested in developing a digital resource to complement and incorporate into their existing school-based boating safety program which usually runs between 40-60 minutes in duration. They commissioned VRTY (<https://vrtty.io/>) an education 360-degree VR provider to develop a proof-of-concept (or first iteration) of a learning resource that could be used on screen (tablet, smart phone, computer) or in a VR headset that had a smart phone inserted into it. A planning day was held with safety officers to scope out key safety messages for the resource based on their existing education program and a boating handbook published by government. The author of this paper was commissioned to provide expert advice of learning design and game mechanics and evaluate the proof-of-concept resource with a select group of primary (elementary) school children.

The 360-degree boating safety resource comprised a title page scene, four educative scenes, and a credit scene at the conclusion. The four interactive educative scenes present a story about a family fishing trip in a linear narrative with educative interactive audio, video, images, and textual safety message markers popping up in the scene through click for screen devices such as smart phones or gaze-based interaction in VR mode. There is also a cartoon character called Penny the Pelican who appears in each scene to act as a navigation portal between scenes. The pelican is a character in existing children's safety resources used by safety officers. Table 1 provides an outline of the resource and its interactive content. Figures 2 and 3 are screenshots of Scenes 2 and 3 of the learning resource.

Table 1. Overview of VR learning resource

<i>360 scene description</i>	<i>Number and types of interactive markers</i>
Scene 1 - At the boat ramp: The opening 360-degree scene begins with a still photo of a boat ramp at a harbour then moves to video once an audio voice-over of the child narrator –	- Three signs on the boat ramp provide audio, text, and photo pop-up information on safe car speed at boat ramps and

<i>360 scene description</i>	<i>Number and types of interactive markers</i>
Sammy – begins. Sammy provides an explanation of how to interact with content and use the animated Penny the Pelican character as a navigation portal. The scene shows the boating safety officer van, actual signs about boating safety and a car reversing a boat into the water.	awareness of rowers using the boat ramp. An adult voice narrates this information. - A marker video pop-up of a boating safety officer explaining their role. - A pop up marker on safety information that is boat-specific - A non-mandatory multiple-choice quiz based on information in scene.
Scene 2 - On the jetty: A scene with the two adult characters – Aunty Jess and Uncle Ming who are taking Sammy fishing – holding life jackets. Sammy narrates why life jackets are important.	- A video of the adult characters putting on their life vests with a picture accompanied by an audio, text, and image pop-up on why life jackets are important. - A child's life jacket at the feet of the adults is a pop-up that has audio, image, and text on the type of life jacket suitable for children.
Scene 3 - Safety gear on the boat: A scene which shows a range of safety gear on the deck of the boat. Sammy explains that the user must interact with all the equipment to find out what it is does.	- There are 11 images of pieces of equipment (markers) on the boat's deck that have an animated sparkle over them indicating that the user should interact with them. Each marker has audio, text, and image. Markers include: a bottle of drinking water; paddles; emergency V sheet; bilge pump; fire extinguisher; flare; electronic distress beacon; waterproof floating torch; bailer bucket; marine radio; anchor; and map. The sparkle disappears after the user clicks on the equipment and the pop up is activated. - A non-mandatory multiple-choice quiz appears after the markers are clicked on.
Scene 4 - On the water: This scene begins with a 360 still image with Sammy explaining awareness while boating is important and that they have been asked to be the boat's 'look out' while Aunty Jess is fishing. There are several text and numbered distance indicators such as 'dive flag', 'swimmer' etc indicating what to look for. The scene then becomes a video with the swimmer swimming etc.	There are two types of marker information conveyed: Firstly, as the video activates the number indicating metres changes as the swimmer or kayak moves closer to the boat. Other distance indicators are for a dive flag and another boat. Audio, image, and text on markers explain mandatory distance. - There is a non-mandatory multiple-choice quiz



Figure 1: Screenshot of scene 2.



Figure 2: Screenshot of Scene 3.

4 STUDY DESIGN AND PARTICIPANTS

In October 2022, the author facilitated a 50-minute focus group using the proof-of-concept learning resource with eight children (4M, 4F aged 10 - 11 years). The children were students at a government primary (elementary) school located in a regional coastal town in NSW, Australia. The purpose of the focus group was to elicit feedback from the prospective end-users of the learning resource (children in Years 5 and 6 of primary school). A focus group methodology was developmentally appropriate for the children who were enthusiastic about taking part, excited about using the VR headset and thoughtful about the oral and written feedback they supplied. The author, who has over two decades of experience in facilitating focus groups including with children and young people, also possesses a good knowledge of child development and research ethics with children [15], which are all consider prerequisites for conducting this type of inquiry [16].

The study was approved by the University of Newcastle human research ethics committee (H-2017-0229) and NSW Department of Education (SERAP 2017396). Prior to the focus group, the principal of the school sent home an information statement and parental/care-giver consent and child assent form to families.

The focus group was conducted in a quiet meeting room at the school. The author facilitated the group, with a staff member from VRTY present to trouble-shoot technical issues and a Maritime Education Officer also attending. Equipment included a smart phone pre-loaded with the 360-degree learning resource and a VR headset that the mobile phone could be inserted into. The headset had headphones. Each child had their own headset and were given paper and pen for other activities. The focus group was audio recorded and conducted according to the schedule outlined in Table 2.

Table 2. Schedule of focus group activities

Time	Activity
5 minutes	Introduction to research and activities. Answering children's questions. Writing of name tags.
2 minutes	Group discussion: How many times have you tried VR before today?
2 minutes	Explanation of VR safety, modes of interaction in the learning resource.

Time	Activity
20 minutes	Playing the learning resource.
10 minutes	Each child fills out "What did I like" and "What could be improved" about the learning resources using sheet supplied. Group discussion of responses.
10 minutes	Systems Useability Scale (child friendly version) administered by researcher and safety officer.
3 minutes	Final group discussion and thank you for participation.

The study used a mixed methodology which included:

1. Audio recording of focus group discussion.
2. Unstructured observations of children using the resource and equipment undertaken the adults present an audio recorded post focus group.
3. Written comments from children using the 'What I like' and 'What could be improved' feedback sheet.
4. The Systems Usability Scale (SUS) [17], child-friendly emoji version of the SUS [18].

Inductive qualitative analysis [19] was undertaken to identify key themes from the audio recording of the focus group and adult observation conversation recorded post focus group, and the children's written responses in the feedback sheet. The SUS [17] is an industry standard 10 item questionnaire with a 5 response Likert scale option for respondents: Strongly Agree, Agree, Neither Agree or Disagree, Disagree and Strongly Disagree. A child-friendly version [18] with the written Likert scale options matched to emojis was administered by the author and the safety officer present (the children were split into two groups). The SUS has been used to evaluate the usability of hardware and software applications. In this evaluation it was used to evaluate what the usability of the learning resource (combination hardware and software) from the children's perspective. Data was analysed using an adjective rating of SUS scores [20].

5 FINDINGS

All participants had some prior experience in using a VR (Table 3) for leisure (e.g., gaming with friends) but not in formal learning contexts. Children were given the option of experiencing the learning resource in 360-degree screen mode on the smart phone or via the VR headset. All children initially chose to experience the learning resource via the VR headset. One child expressed discomfort with this and took the headset off and used screen mode.

Table 3. Participant's prior experience of VR

	Number of times children had used VR prior to focus group.	
	Female	Male
1-5 times	2	4
6-10 times	1	0
More than 10	1	0

Observations and the audio recording indicated that while children enjoyed the learning resource in VR, most complained that it was 'blurry' and some had trouble adjusting the straps of the headset despite adult assistance. It is plausible that in the limited time students had to experience the resource (20 minutes) that there was not enough time to adjust the headsets and that headsets moved around affecting the quality of viewing experience. The 20-minute timeframe realistically reflected the

time that would be given over to use of the resource in a standard education session conducted in schools. A check of the devices after the focus group indicated that only two children managed to experience all four educative scenes with the rest experiencing only two or three scenes.

The written feedback sheet and audio recorded discussion indicated that while the issue of a blurry VR experience was common, children liked the information pop-ups, the animated pelican, and the feeling that the characters of the child narrator and the aunt and uncle were directly speaking to them. They also felt like they were on a fishing trip and liked that they could experience this either on a mobile device screen or in a VR headset. The following comments from the written feedback sheet (FS) and audio recording (AR) reflect some of these themes:

“I liked how it felt like they (the character in the resource) were talking to me (and) the pelican (pelican). I like how you can look at it on the headset and the phone.” (FS)

“I like it when I went to the boat (and) how the signs worked (with information pop-ups).” (FS)

“(To improve it) the straps (on the headset) need to go tighter, you could interact with more items, and (it was) a little blurry sometimes.” (FS)

“(I wanted to interact) with the safety stuff on the boat, like if there was a fire get the bucket with water (to put it out).” (AR)

“It was a bit blurry but then I fixed it. (The author then asks who managed to fix the blurriness and three out of eight children put their hand up).” (AR)

Results from the SUS adjective scale rating (Table 4) indicated children had a positive experience of using the learning resource.

Table 4. Systems Usability Scale results

<i>Adjective rating</i>	<i>Result</i>
Worst imaginable	0
Poor	0
Ok	0
Good	50% (4)
Excellent	25% (2)
Best imaginable	25% (2)

6 DISCUSSION

Several learning affordances of VR lend themselves to the development of a resource to educate children about boating safety. These include the ability of VR to transport the learner to the ‘real’ context (a waterway) in which specialised knowledge can be learnt and applied, and the spatialised nature of the environment where behaviours, such as visual awareness of objects and potential dangers, can be practiced. The children who participated in the focus group pointed to the first-person perspectives affordance when they remarked that it felt as though the characters in the virtual environment were talking directly to them, enhancing engagement with delivery of safety messages. The blending of videos of real people and the animated character of the pelican in the virtual environment also increased enjoyment. The request that more interactive, purely fun components be added to the resource indicated that children could

see the potential for such an application and that having the agency to act in the virtual space was important to them. The SUS results confirmed that the children had a positive experience of the learning resource despite complaints of blurriness when wearing the headset.

While children were positive about the learning resource, most did not manage to navigate through all four educative scenes. This may have been due to the allotted time (20 minutes) which corresponded with the time that could be given over to using the resource during school education sessions. Generally, the children were not experienced users of VR and so may have not been wholly comfortable with wearing and adjusting a headset and interacting in, and navigating between, scenes. Around half fixed the blurriness issue by adjusting the headset themselves; even having adults to assist in this did not resolve adjustment and blurriness issues for most. Affordable smart phone headsets such as the ones used in this study are made for adult sized heads and so addressing this issue will be difficult.

Children liked the choice of using the 360-degree learning resource on a computer or mobile device screen or in a VR headset. It may be that within the time constraints of the boating safety sessions conducted in schools that the learning resource might be best experienced on a computer or mobile device screen mode rather than VR mode so that children can better ‘get through’ the content. The next iteration and evaluation of the resource can comparatively test this assumption. One advantage of the resource is that it can be used on personal laptops or classroom computing devices such as smart boards in schools that cannot be visited by maritime safety officers.

7 CONCLUSION

Understanding when VR might be best used for educational purposes is an ongoing, complex project. This is especially true when designing and implementing VR in educational settings such as school, whether that be teachers integrating VR resources into lesson time or as part of community learning programs carried out by external stakeholders. Both scenarios require VR learning resources that strike a balance between interactivity for fun and for learning, and that garner the perspectives of end-users including children, to do this. The present research highlights both the potential of 360-degree learning resources to engage children and the drawbacks, especially in adjusting headsets for visual quality in time-constrained contexts.

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